BUSINESS REQUIREMENTS DOCUMENT

AI-First CRM – HCP Module

Log Interaction Screen

Version 2.0 · July 2026 · Interview Assignment – Round 1

1. Document Information

Field	Details
Document Title	BRD – AI-First CRM HCP Module: Log Interaction Screen
Version	2.0 (Revised – All Gaps Resolved)
Date	July 2026
Prepared For	Interview Assignment Round 1
Status	Final
Deadline	36 Hours from Assignment Receipt
Source Materials	Assignment .docx + Video Transcript (Instructions Task 1.mp4)

2. Executive Summary

This document outlines the business requirements for an AI-First CRM system for the Life Sciences / Pharmaceutical industry, focused on the HCP Module — specifically the Log Interaction Screen.

Field reps currently spend significant time manually logging HCP visits. This system eliminates manual entry entirely: a LangGraph-powered AI agent driven by Groq LLMs controls all form population and updates through a conversational interface. The user interacts exclusively via the AI Assistant chat panel (text or voice). The structured form is display-only. This principle is absolute per the video instructions.

3. Requirement Source Clarification

Two source materials were provided. They contain one apparent conflict that must be explicitly resolved:

■ Conflict: Assignment Document vs. Video Instructions

Assignment doc says: 'offer users flexibility to log via a structured form OR conversational chat' — implying manual input is allowed. Video says definitively: 'The form must NEVER be filled manually; it must be controlled entirely through the AI assistant.' Resolution: VIDEO INSTRUCTIONS take precedence. 'Structured form' refers to the display format, not a manual input method. All data entry flows exclusively through the AI Assistant chat panel.

Source	Authority Level	Notes
Video Instructions (Task 1.mp4)	PRIMARY — overrides document	Explicit rule: form never filled manually
Assignment Document (.docx)	SECONDARY	Tech stack, tools, submission requirements
Provided Screenshots	REFERENCE	UI layout and field design guidance

4. Background & Problem Statement

4.1 Background

Pharma companies rely on field reps to document HCP engagements. Traditional CRMs require manual form entry — slow, error-prone, and often completed hours after visits when details fade.

4.2 Problem Statement

- Manual data entry disrupts the post-visit workflow of field reps.
- Multi-field forms are tedious; errors require navigating complex edit screens.
- No intelligence layer to infer sentiment or suggest next best actions.
- Existing CRMs are not designed for conversational, AI-first paradigms.

4.3 Proposed Solution

- LEFT panel: structured read-only form — displays AI-logged interaction data.
- RIGHT panel: AI Assistant accepting text or voice input.
- LangGraph agent (Groq gemma2-9b-it primary) processes all input and drives form population.
- Two input modes for logging: chat text and voice note — both feed the Log Interaction tool.

5. Project Objectives

#	Objective	Priority
O1	Split-screen Log Interaction Screen (read-only form + AI chat)	Critical
O2	LangGraph AI agent fully controls form state via NLP	Critical
O3	Log Interaction tool — supports text and voice input paths	Critical
O4	Edit Interaction tool — updates specific fields, preserves others	Critical
O5	3 additional sales-related LangGraph tools	High
O6	All NLP logic driven by Groq LLM — no hardcoded parsing	Critical
O7	Persist interaction records in PostgreSQL via FastAPI	High
O8	React + Redux frontend with Google Inter font	Medium
O9	GitHub repo with README + 10–15 min video covering all 5 tools	High
O10	Voice Note: browser consent → STT → Log Interaction tool	Medium

6. Stakeholders

Role	Description	Involvement
Field Representative (Primary User)	Pharma sales rep logging HCP visits	Daily user — types or speaks interaction summaries
Healthcare Professional (HCP)	Doctor / Specialist being visited	Subject of logged interactions
CRM Administrator	Manages system config and HCP data	Manages HCP list
Sales Manager	Reviews rep performance	Reads logged interactions and outcomes
Interview Panel	Technical evaluators	Reviews GitHub repo, video, and live demo

7. Scope

7.1 In Scope

- Log Interaction Screen with split-screen layout (read-only form + AI chat)
- LangGraph AI agent with exactly 5 tools
- Groq LLM integration: gemma2-9b-it (primary), llama-3.3-70b-versatile (extended context)
- Two input modes: chat text and voice note (both flow into Log Interaction tool)
- React + Redux frontend; FastAPI Python backend with CORS; PostgreSQL database
- GitHub repository with README and 10–15 minute demo video

7.2 Out of Scope

- Other CRM modules, user authentication, mobile app, real HCP database, multi-language support
- Voice Note as a separate LangGraph tool (it is an input method, not a tool)

8. Functional Requirements

8.1 Screen Layout

- Horizontal split-screen: Left Panel (~65%) + Right Panel (~35%), full viewport height.
- Left Panel: structured interaction form — read-only, populated only by the AI agent.
- Right Panel: AI Assistant chat with scrollable thread and input bar at bottom.
- Form fields highlight with a yellow flash animation when AI auto-populates them.

8.2 Form Fields (Left Panel — ALL READ-ONLY)

Field	UI Type	Populated By
HCP Name	Text display (typeahead styling, not editable)	AI Agent — Log Interaction tool
Interaction Type	Dropdown: Meeting / Call / Email / Conference	AI Agent
Date	Date display (read-only)	AI Agent — resolves 'today'/'yesterday'
Time	Time display (read-only)	AI Agent
Attendees	Tag chips (read-only)	AI Agent
Topics Discussed	Textarea (read-only)	AI Agent
Materials Shared	Tag chips + Search/Add button (read-only)	AI Agent
Samples Distributed	Tag chips + Add Sample button (read-only)	AI Agent
HCP Sentiment	Pill buttons: ■ Positive / ■ Neutral / ■ Negative	AI Agent — LLM-inferred from tone
Outcomes	Textarea (read-only)	AI Agent
Follow-up Actions	List (read-only)	AI Agent — Schedule Follow-up tool

8.3 AI Assistant Panel (Right Panel)

- Chat thread: user messages right-aligned (blue), AI responses left-aligned (light blue/green).
- Input field placeholder: 'Describe Interaction...'; bold blue Send button (circular icon).
- AI responses confirm populated fields and offer next steps; typing indicator while processing.
- Initial greeting: 'Log interaction details here (e.g., Met Dr. Smith...) or ask for help.'

8.4 Voice Note Input Flow (NOT a separate tool)

■ Design Decision — Voice Note

Voice Note is NOT a 6th LangGraph tool. It is an alternate input method feeding into the same Log Interaction tool. Pipeline: Browser MediaRecorder API (requires consent) → Raw audio blob → Groq Whisper STT → Transcribed text → Log Interaction tool → Form population.

- 'Summarize from Voice Note (Requires Consent)' link below Topics Discussed — triggers mic consent.
- Audio sent to POST /api/voice/transcribe → Groq Whisper returns text.
- Transcribed text injected into chat input and submitted to log_interaction — identical to typing.

9. LangGraph Agent & Tool Specifications

The LangGraph agent receives user messages, uses the LLM to decide which tool to invoke, executes it, and returns a structured response that updates the Redux store. All extraction logic is LLM-driven — no hardcoded parsing anywhere.

9.1 Agent State Schema

State Key	Type	Description
messages	List[BaseMessage]	Full conversation history (HumanMessage + AIMessage)
interaction_data	Dict	Current form field values extracted so far
tool_called	str | None	Name of the last tool invoked
tool_result	Dict | None	Output from the last tool execution
next_action	str	'tool' | 'respond' | 'end'

9.2 Agent Graph Structure

- Node 1 — llm_node: Calls gemma2-9b-it with conversation + tool schemas; decides tool to call.
- Node 2 — tool_executor_node: Executes the selected tool with LLM-generated arguments.
- Conditional edge: llm_node → tool call detected → tool_executor_node → back to llm_node (loop); else → END.

Tool 1 — log_interaction [MANDATORY]

Tool Name	log_interaction
Input Paths	Path 1: User types in chat | Path 2: Voice → Groq Whisper → transcribed text → same tool
LLM Role	Extract: HCP name, type, date/time (resolve relative dates), attendees, topics, materials, samples, sentiment (INFER from tone), outcomes, follow-ups
Output	Structured JSON → FastAPI → Redux store → form auto-populates
Post-Action	AI confirms populated fields + asks if user wants to schedule follow-up
Persistence	Saves to PostgreSQL via POST /api/interactions

Tool 2 — edit_interaction [MANDATORY]

Tool Name	edit_interaction
Trigger	User corrects a field: 'Change the sentiment to Neutral' / 'The date was yesterday'
LLM Role	Identify ONLY the field(s) to update + new value. All other fields stay exactly as-is.
Output	Partial JSON with ONLY modified fields → merged (not replaced) into Redux store
Key Constraint	Must NOT reset or null out any field not mentioned in the correction
Post-Action	AI confirms: 'Updated [field] to [value]. All other details unchanged.'
Persistence	Updates record via PUT /api/interactions/{id}

Tool 3 — schedule_followup

Tool Name	schedule_followup
Trigger	User asks: 'Schedule a follow-up with Dr. Smith next Monday at 2pm'
LLM Role	Extract HCP name, date/time (resolve relative), purpose of follow-up
Output	Follow-up entry added to follow_up_actions field + PostgreSQL update
Post-Action	AI confirms: 'Follow-up scheduled with [HCP] on [date] at [time].'

Tool 4 — suggest_next_action

Tool Name	suggest_next_action
Trigger	User asks 'What should I do next?' OR proactively after negative/neutral sentiment
LLM Role	Analyzes: current sentiment, topics, HCP name, past interaction history from DB

Model	Uses llama-3.3-70b-versatile for extended context reasoning
Output	Sales-relevant AI recommendation in chat — specific to this HCP, not generic

Tool 5 — search_hcp_history

Tool Name	search_hcp_history
Trigger	User asks: 'What did I last discuss with Dr. Smith?' / 'How many times met Dr. Patel?'
LLM Role	Extracts HCP name → FastAPI queries PostgreSQL → LLM generates readable summary
Model	Uses llama-3.3-70b-versatile for history summarization
Output	Human-readable summary in chat; useful for rep visit prep

10. Non-Functional Requirements

Category	Requirement
Performance	AI agent responds within 5 seconds; loading indicator shown during processing
Reliability	All field parsing through the LLM — zero hardcoded logic
Scalability	FastAPI backend stateless; designed for horizontal scaling
CORS	FastAPI must allow requests from localhost:3000 and localhost:5173
Compliance	Voice Note must display browser consent prompt before any mic access
Auditability	All interactions persisted in PostgreSQL with created_at / updated_at timestamps
Error Resilience	Agent failures return graceful chat messages — never crash the UI
Code Generation	All code AI-generated (Gemini 2.5 Pro or ChatGPT 5.0) per assignment rules

11. Technical Architecture & Stack

Layer	Technology	Notes
Frontend	React 18 + Hooks	Component-based; no manual form interaction
State Management	Redux Toolkit	3 slices: interactionSlice, chatSlice, agentSlice
Styling	CSS Modules / Tailwind	Google Inter font via Google Fonts
Backend	Python 3.11+ + FastAPI	REST API; LangGraph agent host; CORS enabled
AI Agent Framework	LangGraph	Tool orchestration and graph-based execution
LLM — Primary	Groq: llama-3.3-70b-versatile	gemma2-9b-it decommissioned; this is the active substitute
LLM — Secondary	Groq: llama-3.3-70b-versatile	Extended context: history summarization, next action
Speech-to-Text	Groq Whisper API (whisper-large-v3)	Voice Note input path
Database	PostgreSQL 16 (Docker)	JSONB for arrays; MySQL acceptable alternative
ORM	SQLAlchemy + Alembic	DB models and migrations

11.1 System Flow (End-to-End)

- 1. User types (or uses Voice → STT) in the AI Assistant panel.
- 2. React dispatches Redux action + sends POST /api/agent with {message, interaction_data, interaction_id}.
- 3. FastAPI passes request to LangGraph agent graph.
- 4. llm_node calls LLM with full conversation history + tool schemas.
- 5. LLM decides tool → tool_executor_node runs it → result returned to llm_node.
- 6. LLM generates human-readable confirmation message.
- 7. FastAPI returns {fields_to_update, ai_message, interaction_id}.
- 8. Redux merges field updates → React re-renders form in real time.

11.2 Redux Store Shape

Slice	State Keys	Description
interactionSlice	hcp_name, interaction_type, date, time, attendees, topics_discussed, materials_shared, samples_distributed, sentiment, outcomes, follow_up_actions, interaction_id	All form values; updated only by AI agent responses
chatSlice	messages: [{role, content, timestamp}], isLoading: bool	Chat thread history and loading state
agentSlice	lastToolCalled: str, agentError: str | null	Agent execution metadata and error state

11.3 Environment Variables

Variable	Example Value	Required By
GROQ_API_KEY	gsk_xxxx (new token from console.groq.com)	Backend — LangGraph + Whisper STT
DATABASE_URL	postgresql://postgres:postgres@localhost:5432/hcp_crm	Backend — SQLAlchemy
ALLOWED_ORIGINS	http://localhost:3000,http://localhost:5173	Backend — FastAPI CORS
BACKEND_PORT	8000	Backend — Uvicorn server
REACT_APP_API_URL	http://localhost:8000	Frontend — Axios base URL

■ Groq API Key Setup

A NEW Groq API key must be created at https://console.groq.com before running. Do not reuse existing tokens. Add as GROQ_API_KEY in backend/.env. README must include this setup step.

12. Data Model — hcp_interactions Table

Column	Type	Description
id	UUID (PK)	Unique interaction identifier
hcp_name	VARCHAR(255)	Healthcare Professional name
interaction_type	VARCHAR(100)	Meeting / Call / Email / Conference
interaction_date	DATE	Date of the interaction
interaction_time	TIME	Time of the interaction
attendees	TEXT	Comma-separated attendee names
topics_discussed	TEXT	Free-text discussion topics
materials_shared	JSONB	Array of materials/brochures
samples_distributed	JSONB	Array of samples distributed
sentiment	VARCHAR(20)	Positive / Neutral / Negative
outcomes	TEXT	Key outcomes or agreements
follow_up_actions	TEXT	Planned next steps
created_at	TIMESTAMPTZ	Record creation timestamp
updated_at	TIMESTAMPTZ	Last modification timestamp

13. API Endpoints

Method	Endpoint	Description
POST	/api/agent	Main agent endpoint. Body: {message, interaction_data, interaction_id}. Returns: {fields_to_update, ai_message, interaction_id}
POST	/api/voice/transcribe	Accepts audio blob → Groq Whisper → returns {transcribed_text}
POST	/api/interactions	Save new interaction to PostgreSQL
GET	/api/interactions/{id}	Get interaction by ID
PUT	/api/interactions/{id}	Update interaction (edit_interaction tool)
GET	/api/interactions/hcp/{hcp_name}	All past interactions for an HCP
GET	/api/hcps	List HCPs for typeahead search
GET	/api/health	Health check — returns {status: ok}

14. Error Handling Requirements

Scenario	User Message in Chat
Groq API timeout / rate limit	'I'm having trouble connecting to the AI right now. Please try again in a moment.'
LLM can't extract HCP name	'I couldn't identify the HCP name clearly. Could you confirm the doctor's full name?'
Mic permission denied	'Microphone access was denied. Please enable it in your browser settings to use Voice Note.'
Voice transcription fails	'I couldn't transcribe the audio clearly. Please try again or type your details.'
Edit field not recognized	'I'm not sure which field you'd like to update. Could you be more specific?'
Database write fails	'The interaction was processed but could not be saved. Please try submitting again.'
Empty / irrelevant message	'I didn't catch any interaction details. Try: "Met Dr. Smith today, discussed Product X."'

15. UI/UX Requirements

- Font: Google Inter (400/500/600/700) applied to ALL elements — import from Google Fonts.
- Color theme: Deep navy (#1A3C5E) primary, sky blue (#0EA5E9) accent, clinical green (#059669) success.
- Left panel: white bg, form grouped into 3 cards (Interaction Details / Discussion & Materials / Assessment & Outcomes).
- Right panel: #F8FAFC bg, chat bubbles with 16px border-radius; user gradient bubble, AI white with left accent border.
- Sentiment: pill button-group — selected Positive=green, Neutral=amber, Negative=red.
- Form field icons (lucide-react): User, Briefcase, Calendar, Clock, Users, ClipboardList, FileText, Package, Heart, Target, Bell.
- Send button: circular, icon, #1A3C5E bg — no stacked text.
- Yellow flash animation (0.6s) on field when AI populates it; 0.2s transitions everywhere.
- 'No materials added.' / 'No samples added.' text: italic, #94A3B8.
- Layout must replicate provided screenshots as closely as possible.

16. Constraints & Rules

#	Constraint	Source	Impact if Violated
C1	LangGraph + LLM must be used	Assignment Doc	Submission rejected
C2	No hardcoded extraction or parsing logic	Video Instructions	Disqualification
C3	Form never filled manually by the user	Video Instructions	Core requirement failure
C4	Edit Interaction only updates specified fields	Video Instructions	Data integrity failure
C5	Groq gemma2-9b-it specified as primary (now llama-3.3-70b-versatile — model decommissioned)	Assignment Doc	Documented deviation
C6	A NEW Groq API key must be created	Assignment Doc	Non-compliance
C7	All code AI-generated — zero human-written code	Assignment Doc	Disqualification
C8	Exactly 5 LangGraph tools	Assignment Doc + Video	Incomplete submission
C9	Voice Note is an input path, NOT a 6th tool	Design Decision	Scope creep
C10	GitHub repo with clear README.md	Assignment Doc	Incomplete submission
C11	Video walkthrough 10–15 minutes	Assignment Doc	Incomplete submission
C12	Google Inter font throughout UI	Assignment Doc	UI requirement failure
C13	FastAPI CORS configured for React dev server	Technical	App won't run locally

17. Acceptance Criteria

ID	Acceptance Criterion	Verification
AC1	Split-screen UI renders correctly, matching screenshots	Visual inspection
AC2	Chat message auto-populates form fields — no manual input	Demo walkthrough
AC3	Voice Note → consent → audio → STT → log_interaction → form	Live demo
AC4	All 5 LangGraph tools functional and demonstrated in video	Video review
AC5	Edit Interaction updates only the specified field(s)	Test with correction prompt
AC6	Groq LLM used for all inference (verifiable in code)	Code review
AC7	Interaction data persists in PostgreSQL after logging	DB inspection
AC8	GitHub repo publicly accessible with setup README	Reviewer access
AC9	Video covers all 9 required sections (see §18)	Video review
AC10	No hardcoded field extraction in codebase	Code review
AC11	Google Inter font applied throughout	Browser DevTools
AC12	CORS configured; React frontend communicates with FastAPI	Network tab
AC13	Error scenarios return graceful chat messages — no UI crash	Error simulation

18. Video Walkthrough Requirements (10–15 min)

#	Required Section	What to Show
V1	Frontend Walkthrough	Split-screen layout; demonstrate form fields are read-only
V2	Tool 1 — Log via Chat	Type NL summary; show form auto-populating with highlight animation

#	Required Section	What to Show
V3	Tool 1 — Log via Voice	Voice Note: consent → recording → STT → form population
V4	Tool 2 — Edit Interaction	Correct a field via chat; confirm only that field changed
V5	Tool 3 — Schedule Follow-Up	Ask AI to schedule; confirm Follow-up Actions updates
V6	Tool 4 — Suggest Next Action	Ask 'What should I do next?'; show tailored recommendation
V7	Tool 5 — Search HCP History	Ask about past interactions; show AI summary from DB
V8	Code Structure	LangGraph graph, 5 tools, Redux slices, FastAPI routers
V9	Task Understanding Summary	Explain your understanding and key design decisions made

19. Submission Checklist

Submit both items separately via: https://forms.gle/XdvLNBJkbdVDGADM8

19.1 GitHub Repository

- Frontend (React + Redux) and backend (FastAPI + LangGraph + 5 tools)
- SQLAlchemy models + Alembic migration files
- docker-compose.yml for one-command PostgreSQL setup
- README.md: project overview, prerequisites, .env setup, Groq key creation step, run instructions
- .env.example for both frontend and backend (no real secrets committed)
- BRD_AI_CRM_HCP_v2.docx included in repo root

19.2 Video

- 10–15 minutes, covers all 9 sections from §18 (V1–V9)
- All 5 LangGraph tools demonstrated live with real Groq API calls
- Code structure walkthrough included
- Task understanding summary included

— End of Document — Version 2.0